

**Problem 4.3**

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**Solution:**

1. For every string  $s \in S$ , pre-compute Karp-Rabin fingerprints of each prefix/suffix  $x$  of  $s$  and the reverse of  $s$  as  $F(x)$  and  $F(x^R)$ .
2. For every string  $s \in S$ , use a slide window of length  $\ell$  to find whether there is a palindrome within  $s$ .

For each suffix  $x$  of length  $i$  and prefix  $y$  of length  $\ell - i$ ,  $xy$  might be a palindrome if

$$F(x) \cdot b^{\ell-i} + F(y) = F(y^R) \cdot b^i + F(x^R)$$

where  $b$  is the base in Karp-Rabin fingerprints.

3. Sort all prefix  $y$  according to  $F(y^R) \cdot b^i - F(y)$ .

For each suffix  $x$ , compute  $F(x) \cdot b^{\ell-i} - F(x^R)$  and check if there is a corresponding  $y$  using binary search.

4. If there is a possible match, verify if it is a palindrome.

If not, find palindrome by brute force.

5. Repeat step 3 and 4 until the palindrome is found.

Runtime Analysis:

$S$  has at most  $\frac{2N}{\ell}$  strings of length over  $\frac{\ell}{2}$  (otherwise, there won't be any palindrome of length  $\ell$ ), each string has at most  $\ell - 1$  prefix/suffix, total number of prefix/suffix is  $O(N)$ , by using subtraction, all Karp-Rabin fingerprints can be computed in  $O(N)$  time. Step 1 and 2 take  $O(N)$  time, step 3 takes  $O(N \log N)$  time. Let  $p = N^{d+2}$  in fingerprints, the probability of using brute force in step 5 is  $O(\frac{1}{N^d})$ , expected time complexity is  $O(\ell N \times \frac{1}{N^d})$ , which is smaller than  $O(N)$  by setting  $d = 1$ . Total time complexity is expected to be  $O(N \log N)$ .